

Temporal Self-Location in Bounded Cognitive Episodes: Separating Process History from Elapsed-Time Access and Control

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Abstract

Physical time can pass through a cognitive episode without implying that the episode represents its own elapsed position. We distinguish process history from temporal self-location, and temporal access from temporal authority: time information may be obtainable without being relevant to continuation or closure. Retrospective duration can instead be reconstructed from remembered content or event structure. Sleep research serves only as a measurement foundation, showing why objective state/history and subjective report dimensions should be kept separate. We propose an undemonstrated waking 2×2 test crossing temporal access and authority during self-terminated deliberation, with content/event reconstruction treated as a competing route. The key evidence would be authority-dependent temporal sampling and closure effects that survive plausible reconstruction and generic-pressure accounts. The framework does not posit a centralized clock or claim that timing information must be conscious. The proposal remains a theoretical identification framework rather than a report of participant results.

Keywords: temporal self-location; duration judgment; deliberation; temporal access; temporal authority; prospective memory; decision termination

1. INTRODUCTION: “HOW LONG HAS THIS THOUGHT BEEN GOING?”

An ongoing thought can be elaborate. It can accumulate evidence, revisit earlier content, generate alternatives, change confidence, and approach or retreat from a conclusion. Yet a different question can remain unexpectedly difficult: how long has this thought been going?

That question is easy to misread. It does not suggest that thoughts lack duration. Every neural and cognitive process unfolds in physical time. Nor does uncertainty about elapsed time imply that the nervous system lacks timing mechanisms. The narrower problem is representational and functional: the history that produced the current cognitive state need not itself constitute an accessible coordinate locating that state in elapsed time.

This paper calls that coordinate temporal self-location. The phrase denotes usable information that locates an ongoing bounded cognitive episode relative to elapsed time or to a

containing temporal frame. This use is narrower than chronesthesia and mental-time-travel accounts of self-location in subjective time, which concern mentally locating oneself in the past, present, or future (Tulving, 2002; Arzy et al., 2008); here the term refers only to the online, usable location of the current bounded episode relative to elapsed or containing time. The central distinction is therefore simple:

process history $\neq$ temporal self-location.

A process can change repeatedly without maintaining an accessible answer to “how long has this episode been underway?” Conversely, a person can acquire a highly precise elapsed-time estimate from a clock with little knowledge of the internal trajectory that occupied that interval. These are separable informational objects even when they correlate in ordinary circumstances.

The distinction sits between several well-developed literatures without replacing them. Research on duration judgment has long distinguished prospective timing, in which participants know that time matters, from retrospective timing, in which duration is requested unexpectedly and can depend strongly on memory and event structure (Block & Zakay, 1997; Zakay & Block, 2004). Work on event segmentation shows that boundaries and remembered event organization influence later duration judgments and memory updating (Boltz, 1995; Swallow et al., 2009; Yue et al., 2026). Evidence-accumulation models describe how decisions terminate as evidence reaches a criterion, and related work shows that costs of deliberation time or deadlines can alter stopping policy (Ratcliff & McKoon, 2008; Drugowitsch et al., 2012; Miletic & van Maanen, 2019; Kira et al., 2025). Time-based prospective-memory research shows that people strategically inspect clocks when they must act at a target time, and that monitoring changes with task demands and consequences (Oksanen et al., 2014; Mioni & Stablum, 2014; Mioni et al., 2020; Laera et al., 2024). Research on spontaneous thought, meanwhile, describes thought as a dynamic sequence whose transitions can be studied and, more recently, whose episode durations can be modeled from reported thought transitions (Christoff et al., 2016; Kérébel & Sackur, 2026).

None of these findings entails that a self-terminating cognitive episode ordinarily maintains an online representation of its own elapsed position. A duration can be reconstructed after the fact. A decision can become more urgent under a deadline. A clock can be strategically checked when a future target makes time relevant. A stream of thought can have objectively or

subjectively identifiable transitions. The present proposal asks how these facts should be related without collapsing them.

The paper makes three moves. First, it specifies the distinction among physical elapsed time, process history, recoverable history, represented duration, temporal self-location, temporal access, temporal authority, and episode closure. Second, it uses sleep as a measured comparison domain in which objective physiology, transition structure, and reports of experience have already been separated rather than treated as one variable. Sleep is not proposed as an analogue of waking deliberation; its role is methodological and dimensional. Third, it proposes a waking experiment that crosses temporal access with temporal authority in a self-terminated deliberative episode. The purpose is to test when elapsed time becomes sampled and control-relevant rather than to infer temporal representation from clock presence or retrospective report alone.

The resulting claim is intentionally limited: physical time may pass through a bounded cognitive episode without being represented or behaviorally authoritative within that episode. This is a hypothesis about a possible dissociation, not an established law of cognition. Its value depends on whether that dissociation can be identified empirically and whether the proposed test survives simpler accounts.

2. PROCESS HISTORY IS NOT TEMPORAL SELF-LOCATION

2.1 Eight objects that should not be collapsed

Consider a bounded cognitive episode beginning at physical time t_0 and closing at t_1 . Its objective duration is

$$T = t_1 - t_0.$$

T is a property of the episode's existence in physical time. It says nothing by itself about what the system represented during that interval.

Within the interval, the episode follows some trajectory through cognitive states. Let $s(t)$ summarize task-relevant content at time t : accumulated evidence, candidate interpretations, confidence, remembered subevents, or other variables appropriate to the task. The sequence $\{s(t): t_0 \leq t \leq t_1\}$ is the process history. This history can be highly structured even if the system never represents T or $t - t_0$ as such.

A third object is recoverable history: the subset or structure of prior states that can later be reconstructed, recognized, or reported. Recoverable history is not identical to process history because many state changes leave weak or inaccessible traces, while later reconstruction can also compress, group, or infer events.

A fourth object is represented duration. After closure, a participant may estimate the episode's duration as $\hat{T}$. An estimate can be accurate or inaccurate, but accuracy does not specify how the estimate was produced. The estimate may derive from online temporal monitoring, remembered event density, learned task regularities, contextual inference, a queried clock, or combinations of these sources.

Temporal self-location is narrower than represented duration. It refers to an accessible representation that locates the ongoing episode relative to elapsed time or a containing temporal frame while the episode is underway. A precise clock query can create such a representation at a moment even if no comparable representation was maintained immediately before the query. Likewise, a system could possess an endogenous timing signal that affects behavior without that signal being available for explicit report. The present proposal therefore does not equate temporal self-location with consciousness.

Temporal access asks whether usable temporal information can be obtained by the episode or participant. Temporal authority asks a different question: whether elapsed time has legitimate decision-policy relevance for continuing, changing, or terminating the episode. A clock can be accessible yet normatively irrelevant to the current decision. Conversely, elapsed time can matter to a task even when precise external time is inaccessible, forcing the participant to rely on endogenous or indirect cues.

Finally, episode closure is the transition at which the locally bounded process ends or hands control onward. In a deliberation, closure can be operationalized by committing to a choice. Closure is not itself evidence of timing. An episode may close because evidence reached a criterion, because a content-defined goal was satisfied, because an external interruption occurred, or because elapsed time acquired sufficient control weight.

These distinctions can be summarized as follows:

Physical elapsed time: how long the episode exists objectively. Process history: what state transitions actually occur. Recoverable history: what of those transitions can later be

reconstructed. Represented duration: an estimate or representation of how long the episode lasted. Temporal self-location: usable online location of the episode in elapsed/parent-flow time. Temporal access: whether usable temporal information can be obtained. Temporal authority: whether time should affect continuation or closure policy. Episode closure: the transition by which the bounded episode terminates or hands off.

The core mistake this framework is designed to prevent is treating covariance among these quantities as identity.

2.2 Why retrospective duration is not enough

The prospective/retrospective timing literature already supplies a strong reason for caution. When participants know in advance that duration will be judged, attention to time can compete with processing of nontemporal information; when duration is requested only afterward, remembered duration can depend more strongly on incidental memory and contextual change (Block & Zakay, 1997; Zakay & Block, 2004). Segmentation can influence retrospective estimates (Zakay et al., 1994), and event organization can alter both the accuracy and variability of unexpected retrospective duration judgments (Boltz, 1995). Recent naturalistic-event work likewise shows that remembered event boundaries can predict retrospective duration judgments of whole events (Yue et al., 2026).

These findings do not demonstrate the present proposal, but they establish a critical identification problem. A participant who reports $\hat{T} \approx T$ after an episode has not thereby demonstrated an online temporal coordinate during the episode. The estimate may have been reconstructed from remembered event structure. Conversely, a poor retrospective estimate does not prove the absence of all online timing signals; encoding, access, or reporting can fail after the fact.

Accordingly, retrospective duration is useful here only as one member of a larger evidence pattern. The proposed experiment deliberately makes the duration question unexpected and combines it with online query behavior, closure dynamics, and content-history measures. The purpose is not to crown any one measure as a direct readout of temporal self-location.

2.3 Memory order is not automatically elapsed time

An ordered memory can preserve “A happened before B” without encoding “B occurred 7.4 seconds after episode onset.” Event count, transition count, and elapsed duration can correlate

strongly in natural cognition, which makes them easy to conflate. Event-segmentation research demonstrates that boundaries shape memory representations (Swallow et al., 2009), and duration research demonstrates that event structure can influence retrospective time judgment (Boltz, 1995; Yue et al., 2026). The appropriate inference is therefore not that memory is irrelevant to time, but that memory-derived temporal reconstruction is a serious alternative to online temporal indexing.

The empirical problem is to decorrelate these sources enough to ask whether elapsed time contributes information beyond the event trajectory that unfolded within it.

3. BOUNDED AND NESTED COGNITIVE EPISODES

3.1 Local episodes inside a continuing system

Cognition is not naturally partitioned into perfectly discrete units, and this paper does not require that it be. The claim requires only tasks in which a local episode can be operationally bounded well enough to measure onset, internal events, and closure. A deliberation beginning with a presented problem and ending with a committed choice is one such case.

That local episode is nested inside a larger system that continues to change. Sensory processing, physiology, background goals, and environmental events continue while the deliberation unfolds. The term parent-flow time will be used descriptively for the containing physical temporal frame relative to which the local episode has an onset and an elapsed position. Parent-flow is not a proposed neural clock, hidden controller, or new biological mechanism. It is a systems description of nesting.

This nesting creates a concrete question. The local episode clearly has a position in parent-flow time from the experimenter's perspective. Does the episode also carry usable information about that position from its own ongoing perspective? If not continuously, can that information be recruited on demand? And when recruited, does it acquire authority over closure?

3.2 Event-defined progression and time-defined progression

Many episodes have an obvious internal progression that is not itself temporal. Evidence has been accumulated or not; an ambiguity has been resolved or not; candidate explanations have been eliminated or not. Such variables can drive continuation policy. A system may therefore

terminate when a content criterion is satisfied without ever needing a precise answer to how many seconds have passed.

Decision models make this point concrete. Bounded evidence-accumulation frameworks explain choices and response times in terms of noisy evidence reaching a decision criterion (Ratcliff & McKoon, 2008). When waiting itself carries cost, optimal stopping can incorporate time costs and allow decision criteria to change with elapsed time (Drugowitsch et al., 2012). Experimental time pressure and deliberation costs can alter response caution or estimated stopping bounds (Miletić & van Maanen, 2019; Kira et al., 2025). These results establish that elapsed time can matter to closure. They do not establish that it always does.

The distinction proposed here is therefore not between “content-driven” and “time-driven” brains. It is between episodes in which parent-flow time is or is not currently available and policy-relevant. The same cognitive system may occupy either regime depending on task structure.

3.3 Spontaneous thought as neighboring, not decisive, evidence

Dynamic accounts of spontaneous thought emphasize transitions among mental states and the constraints that shape those transitions (Christoff et al., 2016). Kérébel and Sackur (2026) go further by modeling the duration of coherent thought episodes from reported thought transitions using an accumulation-to-threshold approach. This work is highly relevant because it treats thought episodes as measurable dynamic objects rather than as an undifferentiated stream.

But episode duration and episode temporal self-location remain distinct. An observer can measure the interval between reported transitions even if the participant did not represent elapsed parent-flow time within the episode. Conversely, a participant can query a clock without thereby revealing the content trajectory that produced the thought. The present proposal therefore treats spontaneous-thought dynamics as evidence that local thought episodes and their transitions can be operationalized, not as evidence that those episodes possess or lack online temporal indexing.

4. WHAT SLEEP ALREADY LETS US SEPARATE

4.1 Why sleep is a measurement foundation

Sleep is useful here because sleep research routinely combines objective state measurements with immediate or near-immediate reports. That pairing creates a methodological precedent for refusing to collapse physical state history, reported experience, and recoverable content into one construct.

The claim is not that waking deliberation resembles sleep. Sleep and waking cognition differ in physiology, sensory coupling, task demands, memory conditions, and many other respects. The point is more disciplined: sleep studies have already had to measure several dimensions that the waking temporal-self-location problem also needs to keep separate.

Those dimensions include physical dwell time, objective state or transition history, experienced duration, recoverable history, content richness, subjective continuity or grouping, and coupling to external temporal information. The value of sleep is the measurement vocabulary.

4.2 Stable-state anchor: structured experience under altered external coupling

Serial-awakening work demonstrates that structured conscious experience can be sampled during sleep rather than inferred only from a morning report. Siclari et al. (2013) repeatedly awakened participants across the night and asked about the presence and properties of the most recent conscious experience. Importantly for the present purpose, the protocol distinguished several report dimensions, including perceived duration, how far back experience could be recalled, and richness or complexity. These measures varied with sleep stage and were not treated as interchangeable.

Later high-density EEG work similarly paired state-resolved physiology with reports of dreaming and dream content (Siclari et al., 2017). NREM serial-awakening work linked dreaming and recall to local electrophysiological features rather than assuming that a global sleep-stage label uniquely determined experience (Siclari et al., 2018).

The foundation provided by these studies is modest but important. A physiological state can persist through objective time while subjective report dimensions remain separately measurable. This is enough to motivate a waking design that measures objective duration, content history,

retrospective duration, and continuity separately rather than assuming that any one is a proxy for the others.

4.3 Fragmented and unstable sleep: transition history is an independent dimension

Sleep microstructure makes the separation even clearer. The cyclic alternating pattern (CAP) was introduced as a physiological component of NREM sleep associated with recurring arousal-related fluctuations (Terzano et al., 1988), later standardized with explicit scoring rules (Terzano et al., 2002) and developed as a marker of sleep instability (Parrino et al., 2012). CAP and related microstructural measures allow investigators to describe an objectively restless or transition-rich state history that is not captured by coarse sleep-stage labels alone.

This matters conceptually because “how much happened in the underlying state trajectory?” and “what kind of continuity or duration is later experienced or recoverable?” need not be the same question. The sleep literature does not by itself prove a particular mapping between CAP transitions and subjective temporal experience. Indeed, that stronger claim would require appropriately paired measurements. Its contribution here is to establish objective transition history as a measurable axis independent of global dwell time.

A more direct dissociation between dwell time and report richness appears in classic REM-awakening work: Rosenlicht et al. (1994) found that increasing prior REM duration from five to ten minutes did not significantly increase dream-report word count. Report length is not identical to phenomenal richness, and the result should not be overread. It nevertheless illustrates why physical dwell time and amount of recoverable report cannot simply be assumed to scale together.

4.4 Repeated awakenings: a template for dimensional separation

The serial-awakening paradigm is especially useful because it minimizes the distance between physiological observation and subjective report. Siclari et al. (2013) used repeated questionings during the night and separately assessed recent experience, estimated duration, recall reach, and richness. One reported pattern is particularly instructive for the present framework: experience could be described as relatively long in duration while richness and recall-back measures did not simply track that duration. The exact sleep-specific explanation is

not imported into waking cognition. The methodological lesson is that dimensions that ordinary retrospective language might collapse can be measured separately.

The sleep lane therefore supports the following narrow conclusion:

Sleep research shows that physical dwell time, coarse sleep state, objective transition/microstructure history, and subjective/report dimensions such as perceived duration, recall reach, richness, and continuity can be measured separately. Serial-awakening work supports separability among report dimensions and sleep state, while CAP/microstructure work supplies an independent objective axis of transition-rich physiological history. The cited CAP evidence does not by itself establish a one-to-one or paired subjective temporal dissociation with those report dimensions.

Nothing in this conclusion says that sleep proves temporal self-location is absent during waking thought, that sleep and deliberation are equivalent, or that sleep research was designed to test the present hypothesis.

5. FROM SLEEP MEASUREMENT DIMENSIONS BACK TO WAKING COGNITION

The cross-domain handoff can now be stated without analogy. Sleep research demonstrates the feasibility and importance of measuring objective state, transition structure, and report dimensions separately. Waking decision research demonstrates that bounded episodes can terminate according to evidence and that time costs can modify closure policy. Time-based prospective-memory research demonstrates strategic clock monitoring when an action must occur at a target time. Duration research demonstrates that unexpected retrospective estimates can depend on remembered event structure.

The missing identification question is not whether humans can perceive time. They can. It is not whether time pressure affects decisions. It does. It is not whether people check clocks when time matters. They do. The missing question is whether temporal information becomes part of a bounded episode's active continuation policy merely because it is available, or selectively when the task gives elapsed time authority over what should happen next.

This framing also clarifies the original orientation problem. "How long has this thought been going?" may require information that is not identical to the thought's causal history. The thought's current content may be the product of many prior transitions, yet an elapsed-time

coordinate may have to be queried, reconstructed, or otherwise recruited. The empirical task is to distinguish those possibilities without treating introspection as evidence.

6. COMPETING ACCOUNTS OF DURATION KNOWLEDGE

A useful theory should lose cleanly to simpler explanations when they fit the data. Seven alternatives are especially important.

6.1 Ordinary retrospective reconstruction

A participant may construct $\hat{T}$ after closure from remembered context, task regularities, or event structure. Prospective/retrospective timing results make this a default competitor, not an exotic one (Block & Zakay, 1997; Zakay & Block, 2004). Therefore, good retrospective duration accuracy alone cannot identify online indexing.

6.2 Content and event count

If longer episodes contain more evidence samples, more transitions, or richer content, event count itself can provide a duration cue. Segmentation and naturalistic-event studies show that event structure can influence later duration judgments (Zakay et al., 1994; Boltz, 1995; Yue et al., 2026). The proposed study therefore jitters the mapping between content events and physical elapsed time and measures both.

6.3 Generic urgency or reward pressure

Making time costly can increase urgency, arousal, or motivation. Faster responses in a time-authority condition might therefore reflect generic pressure rather than use of an elapsed-time coordinate. The common per-sample acquisition cost is not part of this temporal-authority manipulation: it is identical across all four cells and depends only on taking another evidence sample, not on elapsed time or jitter duration. This competitor cannot be defeated by response time alone. The stronger pattern requires authority-dependent time-specific sampling and the randomized Access $\times$ Authority closure pattern, accompanied by measures of urgency and reward pressure. A conditional relation between a self-initiated query and subsequent closure can add convergent evidence after controlling for evidence state, but it is not independently causal because latent urgency, an incipient stopping decision, or unmeasured content state can influence both query timing and closure.

6.4 Clock salience and cueing

A clock or query button may itself cue speed or make the concept of time salient. If merely providing access changes closure when time is explicitly irrelevant, the proposed access/authority distinction is weakened. The no-authority access condition is therefore not a filler control; it is a direct test of this alternative.

6.5 Working-memory and executive demand

Querying and using time can add load. Differences between conditions may reflect generic executive demand rather than temporal self-location. The design minimizes query complexity, keeps the evidence task identical across cells, records workload and difficulty, and relies on the selective relation between elapsed-time information and closure rather than global performance decrements. Statistical adjustment cannot prove away all load differences, so residual ambiguity must be preserved if load tracks the critical effect.

6.6 Simple deadline control

Deadline-sensitive behavior demonstrates that elapsed time can control action when a deadline makes time relevant. It does not show that unconstrained episodes normally represent elapsed time. Time-based prospective memory makes this boundary especially clear. Participants strategically check clocks when they must execute an action at a specified time (Oksanen et al., 2014; Mioni & Stablum, 2014; Mioni et al., 2020), and consequences can alter monitoring behavior (Laera et al., 2024). Those paradigms are strong neighboring evidence for recruited temporal monitoring, but temporal authority is built into the task by the target time. The proposed design asks what happens when access and authority are crossed independently.

6.7 “Memory is time” overreach

Ordered state history is often temporally informative, but temporal informativeness is not the same as an explicit or accessible temporal coordinate. A sequence can preserve ordinal relations without metric elapsed time. If event-history variables fully explain the proposed study’s timing effects, the stronger temporal-indexing interpretation should be rejected rather than redescribed.

7. TEMPORAL ACCESS VERSUS TEMPORAL AUTHORITY

7.1 Access is availability, not control

Let $X \in \{0,1\}$ denote manipulated temporal access. In the proposed experiment, $X = 1$ means that a participant can obtain a precise elapsed-time reading during the episode through an on-demand query. $X = 0$ means that this precise external readout is unavailable.

This manipulation does not claim to remove all endogenous timing signals. Humans may still estimate time from bodily, attentional, mnemonic, or other internal cues. The manipulation concerns access to a known, precise external temporal coordinate. This limit is important: a null effect in $X = 0$ cannot establish that the brain contained no timing information.

Access alone also does not establish self-location. A query button can exist without being pressed. A clock can be visible without affecting policy. Even a successful clock query demonstrates temporal self-location only at the moment the elapsed-time value becomes usable; it does not establish that a continuously updated internal variable existed beforehand.

7.2 Authority is policy relevance

Let $A \in \{0,1\}$ denote temporal authority. $A = 1$ means that elapsed time has genuine relevance to the task's optimal continuation or closure policy. $A = 0$ means that, conditional on evidence, choice accuracy, and the common evidence-acquisition cost specified below, the task's payoff does not depend directly on elapsed time and faster closure does not buy hidden advantages such as more trials or greater reward rate. Both authority levels use the same small fixed cost for each additional evidence sample. That cost depends on taking another sample, not on how long the sample takes to arrive; $A = 1$ differs by adding a direct elapsed-time-dependent task term on top of this common baseline.

Authority is therefore a property of the task-policy relation, not a psychological feeling. Participants may nevertheless experience urgency or pressure when $A = 1$; those reactions are expected consequences and potential alternative explanations, not the definition of authority.

7.3 Why the crossing matters

The 2×2 crossing creates four logically different regimes:

$X=0, A=0$: no precise external temporal query; time has no task authority.

X=1, A=0: precise temporal information is available; time has no task authority.

X=0, A=1: elapsed time matters, but precise external temporal information is unavailable.

X=1, A=1: elapsed time matters and can be precisely queried.

The critical comparison is not simply “clock versus no clock” or “deadline versus no deadline.” It is whether temporal access changes sampling and closure differently when elapsed time has authority than when it does not.

The intended architecture is weakened if X changes closure robustly under A = 0. It is also weakened if A fails to alter closure policy at all, because then authority has not been behaviorally established. In X = 1 trials, authority should produce a systematic change in query behavior if participants actively recruit precise time to guide control. That change need not be a simple increase in total queries; it could be a shift in query timing toward decision-relevant portions of the episode. The prediction should therefore be stated in terms of systematic authority-related sampling, not only query count.

8. PRIMARY STUDY 1: A 2×2 ACCESS $\times$ AUTHORITY TEST IN SELF-TERMINATED DELIBERATION

8.1 Task overview

Participants perform a sequential binary inference task. Each trial begins a bounded deliberative episode. Evidence arrives in discrete samples, and after each sample the participant can either request another sample or commit to one of two choices. The participant determines when the deliberation ends; there is no forced response after a fixed number of samples in the primary task.

The evidence stream is generated so that each sample has known diagnostic value. Critically, the time between samples is jittered within a prespecified range. This decorrelates, to a useful degree, objective elapsed time from the number of evidence events. Two trials can therefore contain the same number of evidence samples but differ in physical duration, while two trials of similar duration can contain different numbers of samples. This feature is central to the identification strategy because it lets event history and elapsed time enter the analysis separately.

The evidence problem should be difficult enough to create genuine continuation decisions but not so difficult that participants abandon the task or respond randomly. Stimulus parameters,

inter-sample delays, the fixed evidence-acquisition cost, and the authority cost function should be calibrated in pilot work, with the calibration procedure fixed before the confirmatory study. The common per-sample cost should be small enough that additional evidence is sometimes worth acquiring and large enough that indefinite sampling is not optimal; immediate stopping should not be trivially optimal. Its value is fixed across all four Access $\times$ Authority cells and does not vary with elapsed time or inter-sample jitter.

8.2 Temporal-access manipulation

In $X = 1$ blocks, an on-demand TIME query is available throughout the deliberation. Activating it briefly displays exact time elapsed since trial onset. A continuously visible clock is avoided because continuous presentation would confound access with constant exposure and make query behavior unobservable. The query display should be brief and standardized. If the query consumes physical time, that small time overhead is recorded and modeled rather than ignored.

In $X = 0$ blocks, the exact elapsed-time readout is unavailable. The task interface is otherwise matched as closely as possible. Again, this condition removes a precise external query, not endogenous interval timing.

For each $X = 1$ trial, two core sampling measures are recorded:

N_Q = number of temporal queries during the episode;

L_Q = latency or timing of temporal queries relative to episode onset and closure.

L_Q should be represented both by first-query latency and by the distribution of query times where multiple queries occur. Query timing can be normalized by objective trial duration only for descriptive analyses because normalization uses the eventual closure time; causal analyses should preserve absolute elapsed query times.

8.3 Temporal-authority manipulation

In $A = 0$ blocks, reward for a correct decision is independent of response time. Trial count is fixed, and post-trial pacing is yoked so that responding faster does not covertly create more earning opportunities or shorten the session. Every additional evidence sample carries a small fixed acquisition cost, $c_s > 0$, identical in all four conditions. This cost is charged for taking another sample; it does not depend on elapsed physical time, the duration of the jittered inter-

sample interval, or authority condition. Accuracy and evidence use therefore matter, and the expected informational benefit of another sample must be weighed against c_s , creating a finite non-temporal stop/continue tradeoff even when elapsed time has no direct payoff authority.

In $A = 1$ blocks, the same fixed evidence-acquisition cost remains in force, while correct-choice value additionally decreases smoothly with elapsed trial time according to a function participants are taught and trained to understand. One illustrative form is

$$R_{\text{correct}}(T) = R_0 \exp(-\lambda T),$$

with error cost specified separately and λ calibrated so that additional evidence sometimes remains worth sampling but waiting is not free. The precise functional form is not theoretically privileged; it should be chosen for stable experimental control and preregistered. The important property is that $A = 1$ adds a direct elapsed-time-dependent term to the same non-temporal sampling baseline used in $A = 0$, so the marginal value of continuing depends on current elapsed time.

A smooth time cost is preferred to a single hard deadline because the study is not intended to reproduce ordinary deadline following. Time remains policy-relevant throughout the episode rather than only near a terminal boundary. Maximum available stakes, evidence statistics, and the fixed per-sample acquisition cost should be matched across authority conditions; only the $A = 1$ task term should vary directly with elapsed time.

8.4 Formal continuation and closure description

At time t within a trial, let s_t denote the evidence/content state available to the participant. The action set is

$$a_t \in \{\text{continue}, \text{choose 1}, \text{choose 2}, \text{query time}\},$$

where query time is available only when $X = 1$. Episode closure is

$$C = \inf\{t : a_t \in \{\text{choose 1}, \text{choose 2}\}\}.$$

Let n_t denote the number of additional evidence samples requested by time t after the initial sample and let $c_s > 0$ denote the fixed cost of acquiring one such additional sample. A compact task-level description is

$$U_t = V(s_t) - c_s n_t - A g(T),$$

where $V(s_t)$ is expected decision value given the current evidence state and $g(T)$ is the known elapsed-time-dependent authority term. The same c_s applies in every Access $\times$ Authority cell and is independent of T and of the jittered delay between samples. Under $A = 0$, the time-dependent term is absent, so continuation is governed by the expected informational benefit of another sample relative to the fixed non-temporal acquisition cost. Under $A = 1$, the identical baseline tradeoff remains and the elapsed-time-dependent term is added, making optimal continuation additionally depend on where the episode lies in elapsed parent-flow time.

This formalization does not posit a new internal clock. It specifies the external task structure. The empirical question is whether participant behavior shows evidence that the externally defined T has become usable within the ongoing policy.

8.5 Trial structure and surprise duration probe

To preserve the distinction between online monitoring and retrospective reconstruction, participants are not told that duration estimation is a target outcome. The cleanest confirmatory implementation uses one unexpected duration probe per participant on a randomly assigned critical trial, before any previous duration question has alerted the participant to timing as a report demand. Assignment of the critical trial across the four $X \times A$ cells is counterbalanced between participants.

Immediately after the choice on that critical trial, participants estimate elapsed duration $\hat{T}$. The primary signed estimation error is

$$E_T = \ln(\hat{T} / T),$$

and absolute error

$$|E_T|$$

indexes estimation precision on the log scale. The log ratio treats proportional under- and overestimation symmetrically and reduces scale dependence across different trial durations.

After the duration estimate, the critical trial includes sparse checks of cue recognition and used content, content richness, perceived continuity, and explicit time-thought. Plausible demand and alternative-process variables are also recorded where feasible: urgency, workload, arousal, engagement, difficulty, perceived control, and reward pressure. These measures should remain brief enough not to turn the experiment into a questionnaire task.

Confidence is obtained after choices on all trials, and accuracy, objective duration, evidence history, query events, and closure are logged continuously.

8.6 Predictions

The proposal does not require a single omnipotent interaction to count as support. It specifies a hierarchy of convergent predictions.

First, the authority manipulation must work. In every condition, participants face the same finite evidence-based stop/continue tradeoff because another sample can improve decision quality but incurs the fixed non-temporal cost c_s . $A = 1$ should alter the observed continuation/closure policy beyond that common baseline in the direction implied by the time-cost function. Participants should, for example, become willing to terminate at lower evidence strength as elapsed waiting becomes additionally costly, consistent with established evidence that deliberation costs and time pressure can affect stopping behavior (Drugowitsch et al., 2012; Miletić & van Maanen, 2019; Kira et al., 2025). If authority produces no policy change, the main experiment is uninterpretable as a test of authority.

Second, temporal access should have little or no practically meaningful effect on closure under $A = 0$. This is an equivalence claim, not a request to celebrate a nonsignificant p value. A preregistered smallest effect size of interest should define what counts as a practically null access effect when time lacks authority.

Third, within $X = 1$, authority should systematically alter temporal sampling. The most direct signature is a change in N_Q and/or L_Q when $A = 1$ relative to $A = 0$. Time-based prospective-memory work establishes that clock checking can be strategic and sensitive to task structure (Oksanen et al., 2014; Mioni & Stablum, 2014; Mioni et al., 2020; Laera et al., 2024). The new question is whether similar strategic sampling emerges when time becomes relevant to self-terminating deliberation rather than to execution at a specified target time.

Fourth, queried time should provide convergent local evidence under authority. Among $X = 1$ trials containing a query, the elapsed-time value revealed at a query should conditionally predict near-term closure probability when $A = 1$ after controlling for evidence strength, evidence count, trial difficulty, and other prespecified state variables; the same relationship should be absent or materially smaller when $A = 0$. This relation adds information beyond total

query count, but it is endogenous at the self-selected query time. Latent urgency, proximity to an internally favored stopping point, unmeasured content state, or another variable can cause both the query and near-term closure. The query-to-closure relation therefore does not independently identify causal control by the revealed time value. It is supporting evidence to be interpreted alongside the randomized Access $\times$ Authority structure and authority-dependent changes in temporal sampling.

Fifth, unexpected retrospective duration estimates provide secondary convergent evidence. If authority-dependent sampling improves $|E_T|$, that pattern is compatible with stronger temporal access during the episode, but it is not sufficient by itself. If clock looking alone improves estimates regardless of authority, the result supports ordinary cue use rather than the stronger architecture.

8.7 Analysis plan and inferential roles

Measures should be organized by role rather than treated as a flat battery.

Manipulation-validity outcomes:

- closure/continuation policy as a function of authority;
- accuracy and reward consequences needed to verify that participants understood the task;
- evidence-sample use under the common fixed acquisition cost, verifying that the cost produces nontrivial finite sampling without differing across the four cells.

Primary indexing/control outcomes:

- N_Q and L_Q under $X = 1$;
- the Access $\times$ Authority effect on episode closure.

Secondary convergent outcomes:

- the conditional post-query relation between revealed T and near-term closure under $A = 1$ versus $A = 0$, treated as endogenous supporting evidence rather than an independent causal contrast;
- unexpected retrospective duration error E_T and $|E_T|$.

Alternative-account and interpretation measures:

- evidence sample count and evidence strength;

- cue recognition and used-content checks;
- content richness and perceived continuity;
- urgency, workload, arousal, engagement, difficulty, explicit time-thought, perceived control, and reward pressure.

A multilevel hazard or discrete-time survival model is a natural way to analyze closure because continuation decisions recur within trials. The model can estimate the probability of closure at each decision opportunity from evidence state, evidence/sample count, elapsed time, authority, access, and their prespecified interactions, with participant-level random effects. Because c_s is fixed by design, cumulative acquisition cost is determined by sample count rather than estimated as a separate time-varying authority term. Query events can be entered as time-varying predictors, but their within-trial coefficients should be interpreted conditionally rather than as randomized causal effects because query timing is self-selected. A separate count or hurdle model can analyze N_Q , while query timing can be modeled as an event process. The exact model family should be chosen after simulation and pilot diagnostics, not by post hoc fit to the confirmatory data.

Power should likewise be simulation-based. Because the decisive quantities include interaction and within-trial event effects, an arbitrary conventional sample size is not defensible. Pilot estimates should be used to choose a participant/trial combination that provides high power—preferably at least 0.90—for the preregistered Access $\times$ Authority closure contrast and the authority-related query effect while preserving an untouched confirmatory sample.

8.8 Separating time from event history

The inter-sample jitter is not merely procedural noise. It creates the possibility of fitting elapsed time T and content/event history as distinct predictors. The fixed acquisition cost does not vary with jitter: two trials with the same number of acquired samples incur the same sampling cost even if their physical durations differ. If closure appears time-sensitive only because later moments contain more evidence samples, the stronger interpretation should fail after evidence count and evidence strength are modeled. Likewise, if retrospective estimates are explained by remembered sample count or event structure, that finding favors reconstruction.

No statistical model can prove that every conceivable content variable has been controlled. The study therefore aims for identification against plausible, measured competitors rather than metaphysical proof of a pure temporal variable.

9. MEASUREMENT AND IDENTIFICATION LOGIC

9.1 What counts as evidence for temporal self-location?

The paper deliberately rejects three shortcuts.

Clock looking alone is insufficient. A participant might query out of curiosity, salience, compliance, or habit.

Retrospective duration accuracy alone is insufficient. Duration can be reconstructed from events and memory.

Closure-time change alone is insufficient. Urgency, reward pressure, evidence dynamics, or generic motor preparation can change response time.

The stronger inference requires convergence. When elapsed time has authority, participants should selectively recruit temporal information, and the randomized Access $\times$ Authority structure should produce the corresponding change in continuation/closure policy. A conditional relation between self-initiated queries, revealed elapsed time, and subsequent closure can support that interpretation, but it cannot independently establish causal control by the revealed value because the local query time is endogenous. Unexpected retrospective duration can then serve as a secondary diagnostic of what was available, not as a privileged readout of online representation.

9.2 Access does not imply continuous maintenance

The proposed study can identify moments at which elapsed time is explicitly acquired and becomes policy-relevant. It cannot, without additional methods, establish whether an internal temporal coordinate is continuously maintained between queries. A query may trigger just-in-time construction of temporal self-location. That possibility is not a rescue hypothesis; it is one of the scientific ambiguities the design intentionally preserves.

Likewise, $X = 0$ does not mean temporal information is literally absent from the nervous system. It means a precise external elapsed-time coordinate cannot be queried. If participants still

behave as if they know T accurately, that result becomes informative about endogenous or inferential sources, but attributing those sources would require a separate study.

9.3 Authority does not require conscious time thought

A participant need not verbally think “seven seconds have passed” for elapsed time to affect policy. The proposed construct is functional: temporally specific information has control weight. Explicit time-thought reports are therefore useful but not constitutive. A dissociation between behavior and explicit report would be scientifically interesting, but it would not be interpreted here as a theory of unconscious timing.

9.4 Relation to established time monitoring

Time-based prospective-memory paradigms provide perhaps the closest neighboring evidence. Participants can inspect clocks strategically, with checking increasing as a target time approaches; monitoring behavior depends on clock availability and can respond to consequences (Oksanen et al., 2014; Mioni & Stablum, 2014; Mioni et al., 2020; Laera et al., 2024).

Two studies are unusually close component precedents. Waldum and Sahakyan (2013) used a time-based prospective-memory task in which participants were to respond after ten minutes while background-song structure varied; their Experiment 2 included no-clock, low-consequence clock, and high-consequence clock conditions. Participants in the high-consequence condition were told that each clock check would extend the ongoing task, and clock checking was substantially reduced relative to the low-consequence condition. Song structure also influenced no-clock target timing and the timing of clock checks when a clock was available, supporting a role for event memory together with external clock information in prospective timing. Huang et al. (2014) likewise examined an eleven-minute time-based prospective-memory task while manipulating reminders and whether clock checking was discouraged. Discouraging clock checking reduced checking and increased ongoing-task costs, which the authors interpreted as greater reliance on internal control; standard clock checking and reminders provided routes for externalizing control.

These studies therefore already establish important neighboring machinery: clock access can be strategically used, consequences or instructions can alter checking strategy, event structure can contribute to prospective timing, and control can shift between more externalized and more

internalized forms. That overlap narrows the present novelty claim. Neither study independently crosses precise temporal access with temporal policy authority inside a self-terminated deliberative episode while making process history versus online within-episode temporal self-location the explicit decomposition and treating content/event reconstruction as a competing route. The present access $\times$ authority crossing asks when elapsed time is recruited into continuation/termination policy, not whether clock monitoring, checking costs, event-based timing, or internalized/externalized temporal control exist. This is an integrated identification extension, not a claim to have discovered those component processes.

10. FAILURE CONDITIONS AND RESCUE RESISTANCE

A useful empirical proposal must specify what would count against it before data arrive. Six outcomes are especially damaging to the intended interpretation.

10.1 Access changes closure when time has no authority

If $X = 1$ reliably changes closure under $A = 0$ by more than the preregistered smallest effect of interest, then the clock/query manipulation is not behaviorally neutral when time is irrelevant. Clock salience, generic cueing, or interface differences become sufficient explanations. The intended access/authority dissociation fails or requires redesign; it should not be relabeled as evidence for ubiquitous temporal control.

10.2 Authority fails to change policy

If $A = 1$ does not alter continuation/closure in the direction required by the payoff structure, temporal authority has not been behaviorally established. Null query effects in that case say little about indexing because participants may not have adopted the intended utility structure.

10.3 No authority-related change in temporal sampling

If N_Q and L_Q show no systematic difference between authority conditions when time is accessible, evidence for active recruitment of temporal self-location is weakened. A closure effect alone could then be produced by generic urgency, implicit timing, or other mechanisms.

10.4 Content/event memory fully explains timing effects

If elapsed-time predictors lose their explanatory contribution once evidence count, evidence strength, remembered events, or other content-history measures are included, then the stronger temporal-indexing interpretation is not identified. Event-history reconstruction wins for that dataset.

10.5 Only clock looking improves duration estimates

If querying improves $|E_T|$ similarly whether or not time has authority, while closure remains unaffected, the result supports ordinary use of an external cue. It does not show that temporal information acquired authority within the episode's control policy.

10.6 No access/authority distinction survives

If all measured consequences of X and A reduce to a single undifferentiated time-salience or pressure factor, the central operational separation fails. The paper should not rescue that outcome by positing hidden timing variables that were never measured.

These failure conditions make the proposal asymmetric in a productive way. Positive evidence must be convergent; clean negative or confounded patterns are allowed to weaken the architecture.

11. DOWNSTREAM FUNCTIONAL-ROLE TRACK: REPORT → RECOMMEND → CONTROL

The primary experiment asks when temporal information is accessed and gains authority during control of an ongoing episode. A downstream functional-role progression can test a related but distinct hierarchy without replacing Study 1.

REPORT asks whether temporal information is available for explicit report. The unexpected $\hat{T}$ probe already provides one minimal instance, though richer report tasks could be developed later.

RECOMMEND asks whether temporal information can guide an explicit judgment about what should happen next without directly controlling the participant's own ongoing episode. For example, a participant could observe a preserved state snapshot from another trial and

recommend “continue” or “stop” when elapsed time is or is not supplied. Such a result would show policy-relevant knowledge at the level of recommendation.

CONTROL asks whether temporal information actually changes the participant’s own continuation or closure behavior. Study 1 targets this level through authority-dependent query use and time-sensitive closure.

The progression should not be read as a developmental sequence or a claim that report must precede control. Its purpose is functional decomposition: information can be reportable without changing action, useful for advice without governing one’s own current behavior, or behaviorally controlling. The ordering remains a downstream program because the access $\times$ authority experiment is the primary identification test.

12. SYSTEMS SIGNIFICANCE

The proposal matters at a systems level because nested processes often possess different state variables and different control-relevant information. An outer process can know something about a subsystem that the subsystem does not currently use. An experimenter knows the elapsed duration of a trial continuously. The local deliberation may not. Treating physical passage, internal transition history, temporal representation, and control weight as one variable obscures that possibility.

The distinction also helps discipline language about subjective time. “The episode lasted ten seconds,” “the participant remembered many events,” “the participant estimated ten seconds,” “the participant knew six seconds had elapsed while deciding,” and “the participant stopped because six seconds had elapsed” are five different claims. The first is physical, the second mnemonic, the third report-level, the fourth representational/access-level, and the fifth control-level. A mature account should be able to ask which relation is supported rather than moving silently among them.

This decomposition is compatible with many mechanisms. Temporal information might arise from interval-timing processes, accumulated state change, memory, external clocks, learned context, or combinations. Control might be implemented through collapsing bounds, urgency signals, value computations, or other architectures. The current paper does not adjudicate among those mechanisms except where they generate distinguishable predictions in Study 1.

The proposed contribution is consequently narrower than a theory of subjective time. It is an identification framework for asking whether a locally bounded episode has an online temporal location and whether that location has authority over closure.

13. SCOPE AND LIMITATIONS

First, bounded episodes are experimental constructions. Real cognition contains overlapping processes, interruptions, and ambiguous boundaries. The proposal requires operationally bounded tasks, not a claim that all natural thoughts have crisp starts and stops.

Second, temporal self-location is defined functionally, not phenomenologically. The experiment can show that elapsed-time information was sampled and influenced policy without determining what that information “felt like.” Conversely, explicit phenomenological reports cannot by themselves prove policy relevance.

Third, the access manipulation concerns a precise external temporal query. It does not eliminate endogenous timing mechanisms. Any inference about the absence of internal time signals would exceed the design.

Fourth, temporal authority is deliberately manipulated through decision consequences. This may increase urgency, arousal, and reward pressure. The randomized Access $\times$ Authority structure and authority-dependent sampling measures provide the main leverage against those alternatives. The local query-to-closure analysis is convergent only because self-initiated query timing is endogenous, and no combination of these measures can guarantee perfect separation. If generic pressure predicts the critical effects as well as temporally specific information, the ambiguity should remain unresolved.

Fifth, retrospective duration remains multiply determined. Even an unexpected probe can be reconstructed from event history. That is why $\hat{T}$ is secondary rather than decisive.

Sixth, the sleep comparison has strict limits. Serial-awakening work supplies examples and methods for separating sleep state from distinct subjective/report dimensions, while CAP/microstructure work supplies an independent objective transition-history axis. The cited CAP evidence does not establish a paired subjective temporal dissociation with those report dimensions. None of this implies that waking deliberation is sleep-like, that hyperfocus is a sleep state, or that sleep proves the waking hypothesis.

Seventh, the proposal makes no claim about a centralized clock variable. Multiple timing processes may coexist at different scales and may influence behavior without conscious access.

Eighth, this paper is not a clinical ADHD account. The hypothesized dissociation, if real, must stand in general cognition without ADHD-specific assumptions. Clinical or hyperfocus applications would require separate evidence.

Ninth, the proposed study identifies temporal recruitment under an experimentally specified authority relation. It does not establish that temporal indexing is absent in ordinary unconstrained thought. A positive result would show conditional recruitment; a broader ecological claim would require additional paradigms.

Finally, the novelty posture must remain conservative. Duration estimation, event-based reconstruction, evidence accumulation, time pressure, clock monitoring, serial awakenings, sleep-state transitions, and spontaneous-thought dynamics all have established literatures. The proposed integrative contribution is the explicit separation of process history from temporal self-location inside bounded episodes, the use of sleep as a measurement foundation for keeping objective and reported histories apart, and the independent manipulation of temporal access and temporal authority in a waking self-termination task. That contribution remains proposed pending empirical test and continued prior-art scrutiny.

14. CONCLUSION

A process can have a history without having an accessible coordinate for where it is in that history's elapsed time. That distinction is easy to lose because physical duration, event count, memory, retrospective estimate, clock access, and stopping behavior often correlate. The present framework keeps them apart.

Sleep research shows why dimensional separation is empirically tractable: serial-awakening work pairs sleep-state physiology with distinct reports of duration, recall reach, richness, and related properties, while CAP/microstructure work supplies an independent objective axis of transition-rich physiological history. Waking decision research shows that bounded episodes can terminate according to evidence and that time costs can change stopping policy. Prospective-memory work shows that people strategically recruit clock information when a temporal target

gives that information clear authority. Retrospective timing work shows why accurate duration reports cannot be assumed to reveal online timing.

The proposed experiment joins those lessons in a deliberately narrow test. Temporal access and temporal authority are crossed during self-terminated deliberation. The design asks whether precise temporal information is sampled and affects closure specifically when elapsed time matters to the continuation policy, while physical time is partially decorrelated from content/event count. The key evidence is a convergent pattern across query behavior, closure, content history, and unexpected retrospective report—not any single timing measure.

If access changes behavior even when time is irrelevant, if authority does not change policy, if temporal sampling does not respond to authority, or if event history fully explains the effect, the stronger proposal should weaken. No hidden timer is needed to rescue it.

The empirical status therefore remains straightforward: the distinction is theoretically motivated and testable, but undemonstrated. The question is not whether a thought occupies time. It is whether the thought, while it is happening, knows where it is in that time—and whether knowing matters for what it does next.

Generative AI Use

OpenAI ChatGPT (GPT-5.6 Sol) was used during the 2026 research and manuscript-development workflow for conceptual formalization, literature discovery and organization, adversarial critique and stress-testing, drafting and editing, notation refinement, simulated peer-review workflows, and submission-package preparation. Model outputs were reviewed and adjudicated by the author, who retained intellectual control and final scientific judgment and verified cited sources and factual claims used in the manuscript. Generative AI is not an author and generated no participant data or empirical dataset.

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